

Chapter 1

The Solow Model

Empirical regularities of developed countries, in particular the United States, are as follows.

Fact 1. *There is secular growth in per-capita output at roughly a constant rate. Growth does not diminish over time.*

Fact 2. *Physical capital per worker grows over time.*

Fact 3. *The rate of return for capital is approximately constant.*

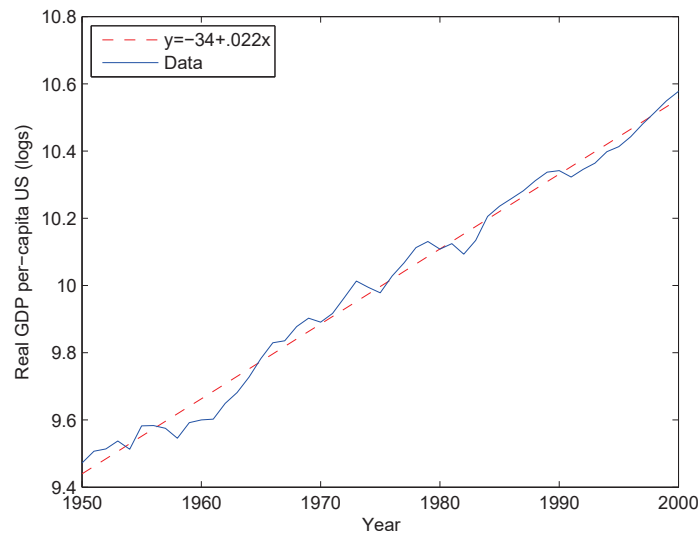
Fact 4. *The capital to output ratio is constant over time.*

Fact 5. *Both capital and labor shares are constant over time.*

These regularities are known as the *Kaldor facts*, following Kaldor (1961).

The first fact is plotted in figure 1.1. Since real income per-capita is in logs, the slope of the trend line (dashed) tells us the growth rate, which is just above %2. The data pretty much stays very close to this trend line, which tells us that the growth rate is constant, ignoring business cycle fluctuations.

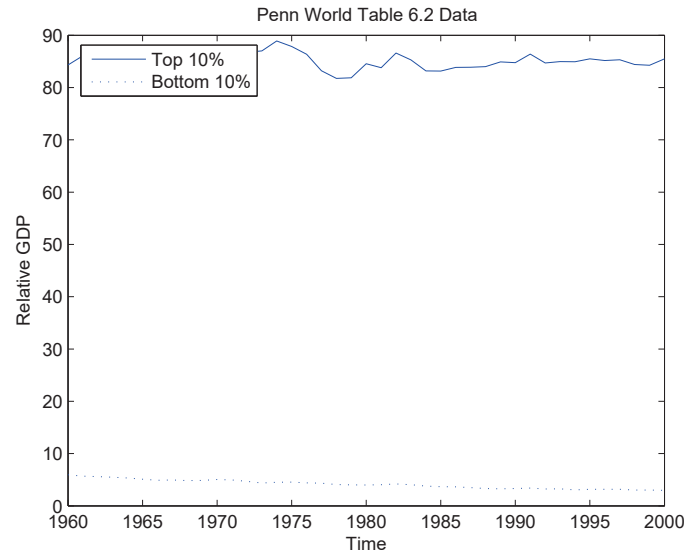
Figure 1.1: Real income per-capita US



A related, important fact which has gained substantial attention is that there are huge differences in GDP per-worker across countries. This fact is displayed in figure 1.2. This figure plots the average GDP per-worker, relative to the 2000 US value, in the top %10 of the world income distribution, along with the bottom %10. On average, the top %10 are 17 times richer than the bottom %10.

At a minimum, any sensible theory of economic growth must be able to generate

Figure 1.2: Cross-country Income Inequality



the first 5 facts. We begin with a basic model due to Solow (1956). The Solow model is able to capture all the facts described above while being very simple. The model describes the dynamics of capital and output with a simple first-order difference equation, by assuming that each period, the household sector saves a constant fraction of income. The model is important because it captures multiple, key features of economic growth that are generated by models with deeper micro-foundations. Therefore, it is very helpful at this stage for intuitive purposes.

1.1 Model

1.1.1 Technology

Let Y denote aggregate output, K the aggregate stock of capital, L the aggregate quantity of labor, and A the efficiency of labor. AL is effective labor and grows over time. The aggregate production function relates inputs to outputs, and is given by

$$Y = F(K, AL), \tag{1.1}$$

Assumption 1. $F : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ satisfies the following conditions,

1. it displays constant returns to scale (CRS)¹,
2. it is twice differentiable over its whole domain,
3. it is strictly increasing in both arguments,
4. it displays diminishing marginal products, and
5. it satisfies the Inada conditions; that is, for $x \in \{K, AL\}$:

$$(a) \lim_{x \rightarrow 0} F_x(K, AL) = \infty,$$

$$(b) \lim_{x \rightarrow \infty} F_x(K, AL) = 0, \text{ and}$$

$$(c) F(0, 0) = 0.$$

¹CRS is equivalent to homogeneity of degree one. Recall that function $F : \mathbb{R}^N \rightarrow \mathbb{R}$ is homogeneous of degree k if for any $\lambda > 0$, $F(\lambda x_1, \dots, \lambda x_N) = \lambda^k F(x_1, \dots, x_N)$.

It is useful to normalize variables by labor in efficiency units. Define $y = Y/AL$, and $k = K/AL$. By the homogeneity property imposed on F , we may define a function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ by $f(k) = F(K/AL, 1)$. This form of the technology will be referred to as *intensive form* of the production function, and it should be noted that $y = f(k)$.

Exercise 1. *Show that Assumption 1 implies that the intensive form production function f has the following properties:*

1. *f is increasing; $f'(k) > 0$ for all k ,*
2. *f is strictly concave; $f''(k) < 0$ for all k , and*
3. *Satisfies Inada conditions: $\lim_{k \rightarrow 0} f'(k) = \infty$, $\lim_{k \rightarrow \infty} f'(k) = 0$, and $f(0) = 0$.*

Exercise 2. *Consider the Cobb-Douglas production function $F(K, L) = K^\alpha L^{1-\alpha}$ for $\alpha \in (0, 1)$.*

1. *Show that this function satisfies all of the conditions from Assumption 1.*
2. *Write this in intensive form and verify that it satisfies the conditions in Exercise 1.*
3. *Consider the production function*

$$F(K, L) = \frac{KL}{K + L}$$

Find the production function in intensive form. What condition fails for this case?

1.1.2 Structure

Time is discrete and runs from $t = 0, 1, 2, \dots$. Aggregate labor L_t grows exogenously at the rate n , so that $L_t = L_0(1+n)^t$. Similarly, the efficiency of labor A_t grows at the rate g , so that $A_t = A_0(1+g)^t$. Denote aggregate savings at t by

$$S_t = sY_t, \tag{1.2}$$

where $s \in (0, 1)$ is the *exogenous* saving rate. Capital accumulates through investment according to

$$K_{t+1} = K_t - \delta K_t + I_t, \tag{1.3}$$

where $\delta \in [0, 1]$ is the rate at which capital depreciates, and I_t is aggregate investment in new capital. We close the model by imposing the aggregate resource constraint

$$Y_t = C_t + I_t. \tag{1.4}$$

Since the economy is closed, $S_t = Y_t - C_t$, and we obtain an equivalent resource constraint

$$S_t = I_t. \tag{1.5}$$

Combining equations (1.2), (1.3), and (1.5), and making use of the intensive form of the production function, one arrives at

$$\underbrace{s A_t L_t f(k_t)}_{Y_t} = (K_{t+1} - K_t) + \delta K_t.$$

Now use the fact that $\frac{K_{t+1}}{A_t L_t} = \frac{K_{t+1}}{A_{t+1} L_{t+1}} \frac{A_{t+1} L_{t+1}}{A_t L_t} = (1+n)(1+g)k_{t+1}$ to arrive at the law of motion for capital

$$k_{t+1} = \frac{s f(k_t) + (1 - \delta)k_t}{(1 + n)(1 + g)}. \quad (1.6)$$

The right hand side of (1.6) can be treated as a function $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, so that $k_{t+1} = h(k_t)$. Hence, h describes the law of motion for capital per-effective worker. That is, given k_t , we can readily calculate k_{t+1} and so on.

Note that h inherits the important properties of f ; that is, $h(0) = 0, h'(k) > 0, h''(k) < 0$, and $\lim_{k \rightarrow 0} h'(k) = \infty$. However, $\lim_{k \rightarrow \infty} h'(k) > 0$, but is still finite which will be all that is needed. See Figure 1.3.

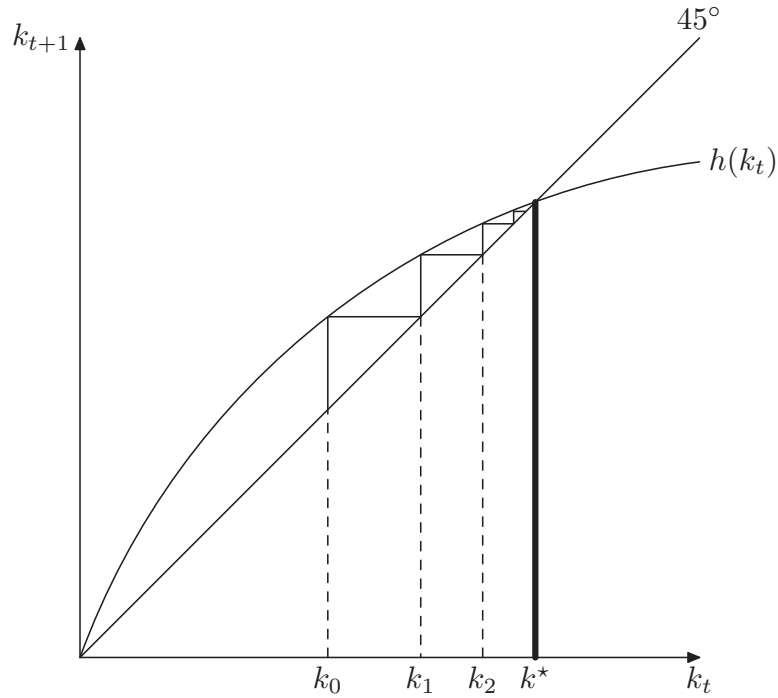
Exercise 3. Show that $\lim_{k \rightarrow \infty} h'(k) < 1$.

1.1.3 Interpretation

An interesting economic interpretation of the law of motion for capital can be constructed after subtracting k_t from both sides of (1.6). Doing this, one arrives at

$$k_{t+1} - k_t = \frac{\overbrace{s f(k_t)}^{T1} - \overbrace{[(1+n)(1+g) - (1-\delta)]k_t}^{T2}}{(1+n)(1+g)}. \quad (1.7)$$

Figure 1.3: Law of motion for capital per-effective worker



The left hand side of this expression is the change in capital per-effective worker between two consecutive points in time. The right hand side consists of two components, $T1$ and $T2$. $T1$ is merely the amount of resources saved (per-effective worker) at t . $T2$, then, is the amount of resources that must be saved (per-effective worker) in order to keep the capital stock constant. Note that $T2$ is being pushed up due to the growing workforce (L), and the growth in efficiency (A), and from natural depreciation of capital. Clearly, when $T1 = T2$, then the capital stock does not change. If $T1 > T2$, then more goods are saved than what is required to maintain the current capital stock, and capital stock tomorrow will be higher. The converse holds when $T1 < T2$.

1.2 Steady-States

Definition 1. *A steady-state is a level of capital per-effective worker k^* , such that $k^* = h(k^*)$.*

In words, a steady-state is a situation in which the stock of capital per-effective worker remains constant over time. In order for this situation to occur, investment in capital per-effective worker must equal the depreciated capital per-effective worker at each point in time. That is, $T1 = T2$ in equation (1.7).

In the basic Solow model described above, there are two steady-states: one (uninteresting) with $k^* = 0$ and another with $k^* > 0$. The steady-state with $k^* = 0$ is sustainable since if the economy starts with no capital, then nothing can be produced, and hence nothing can be invested, and similarly nothing will depreciate so the next period will also trivially have $k^* = 0$.

The case of $k^* > 0$ is more interesting, and given the primitives of the model, convergence to this steady-state will be monotone; that is, if $k_0 > k^*$, then $k_t \searrow k^*$, and if $k_0 < k^*$, then $k_t \nearrow k^*$. This is easily seen from Figure 1.3. The existence of a positive k^* comes from the fact that $\lim_{k \rightarrow 0} h'(k) = \infty$ and $\lim_{k \rightarrow \infty} h'(k) < 1$, along with $h(0) = 0$ and continuity of h . Uniqueness follows by strict concavity of h .

Finally, the monotonicity of h delivers monotone convergence to the steady-state. Figure 1.3 displays convergence starting from $k_0 < k^*$, while the dynamics when starting

from $k_0 > k^*$ are omitted but similar except $\{k_t\}$ would be a decreasing sequence in that case.

Example 1. Let $Y = K^\alpha(AL)^{1-\alpha}$. Then it is easy to check that the intensive form production function is $f(k) = k^\alpha$. It then follows that

$$k_{t+1} = \frac{sk_t^\alpha + (1 - \delta)k_t}{(1 + n)(1 + g)}$$

from equation (1.6). To solve for the steady-state, write

$$\begin{aligned} k &= \frac{sk^\alpha + (1 - \delta)k}{(1 + n)(1 + g)} \\ [(1 + n)(1 + g) - 1 + \delta]k &= sk^\alpha \\ k^* &= \left[\frac{s}{(1 + n)(1 + g) - 1 + \delta} \right]^{\frac{1}{1-\alpha}}. \end{aligned} \quad (1.8)$$

1.3 Balanced Growth

Closely related to the concept of a steady-state is a *balanced growth path*.

Definition 2. *Balanced growth is a situation in which all aggregate variables grow at a constant rate. A balanced growth path (BGP) is a trajectory that satisfies balanced growth.*

In the Solow model, note that when k_t is constant over time, then $K_t = A_t L_t k^* = [(1 + n)(1 + g)]^t k^*$ and evidently must grow at the rate $(1 + n)(1 + g)$. Hence, a steady-

state for k corresponds to a balanced growth path for K . Similarly, aggregate output $Y_t = A_t L_t f(k^*) = [(1+n)(1+g)]^t f(k^*)$ and also grows at the rate $(1+n)(1+g)$.

In practice we are mostly interested in output per-worker and capital per-worker, instead of per-effective worker. So define $\tilde{y} = Y/L$ and $\tilde{k} = K/L$ to be precisely that. Then the growth rates of both of these objects along the BGP is $(1+g)$, the rate at which efficiency is growing.

1.4 Markets for Inputs

So far we have been silent as to how the labor and capital markets work. In a world where firms hire factors (K and L) and households supply them, we must consider a price system which allocates these factors, and say something about the supply and demand for these factors.

That situation is not fully specified here (we will do it later) but a few things can be said at this point regarding rental prices for capital and labor. We will assume that both input markets are perfectly competitive; that is, both firms and households are price takers, and all inputs are homogeneous. In particular, we will suppose that households supply both factors inelastically. Firms operate the CRS technology. Capital is rented at a rental rate R , and labor in efficiency units is paid the wage w . The actual interest rate r is not to be confused with the rental rate R , but this will come later.

With this setup, we may with no loss in generality assume that there is one represen-

tative firm which behaves competitively. The profit a firm earns is $ALf(k) - wAL - RK$.

Exercise 4. *Show that profit maximization implies that $R = f'(k)$ and $w = f(k) - kf'(k)$, and that the aggregate firm makes zero profits (Hint: for the second part, use Euler's Theorem for homogeneous functions).*

To map some of the models implications to the data we define *Capital's share in output* as $\frac{\text{payments to capital}}{\text{value of output}}$, and *effective labor's share in output* as $\frac{\text{payments to effective labor}}{\text{value of output}}$.

Combining these definitions with our notation, capital's share becomes RK/Y , while effective labor's share becomes wAL/Y . It is easy to see that both of these objects are constant along a BGP.

Exercise 5. *Show that when $Y = K^\alpha(AL)^{1-\alpha}$, then capital's share is α and labor's share is $1 - \alpha$.*

It should be noted that the effective wage w is in efficiency units. However, the wage per unit of L (wage per worker), which we denote by W is equal to wA , the total payments to raw or individual labor. So clearly, payments to raw labor grow at the rate $(1 + g)$ along the BGP.

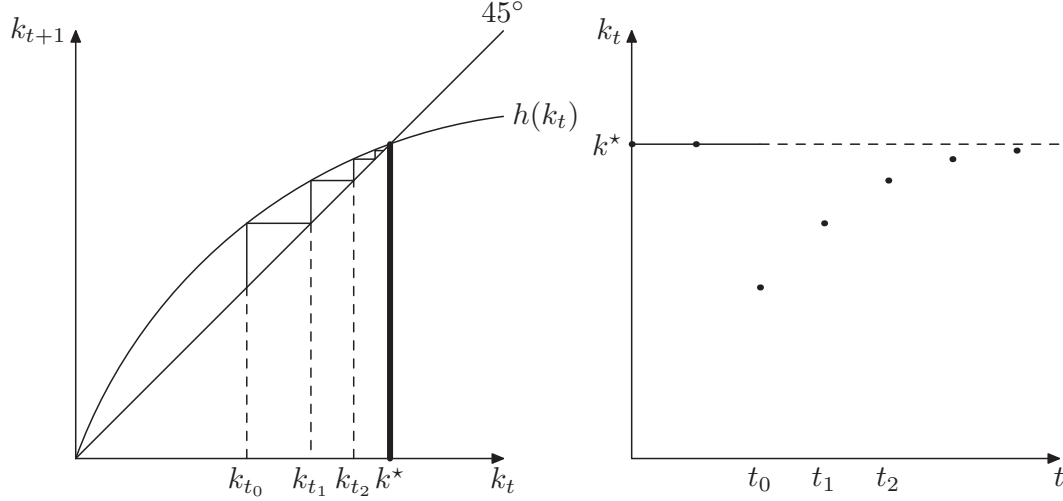
1.5 Transitional Dynamics

Although we make use of the idea of steady-state, there is no reason to believe that an economy is always at such a point. For example, if the initial capital per-effective

worker is different from this value, there is a certain path the economy will follow as it approaches a steady-state. Similarly, if the economy is in steady-state, major events can occur to push it out. We will now consider different cases for expositional purposes, where we perturb a steady-state with an exogenous change in the environment.

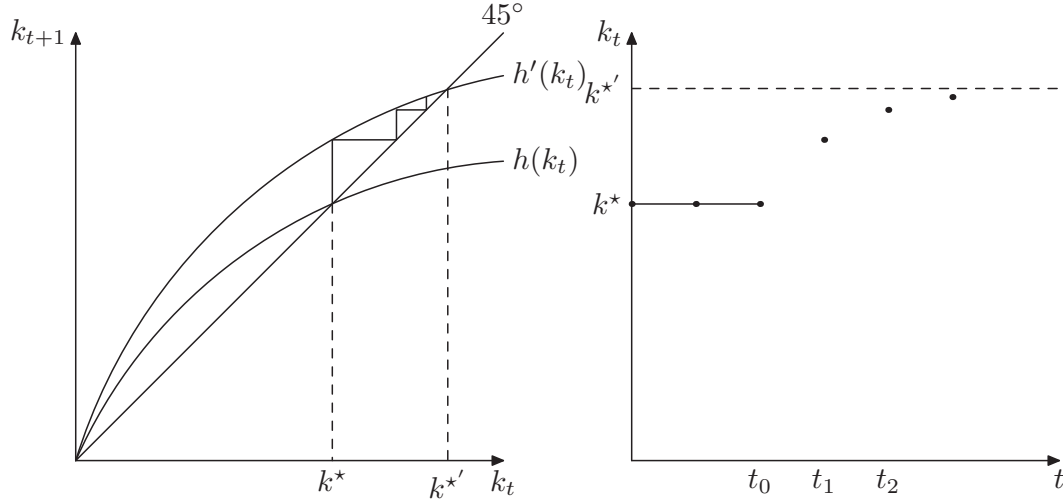
An Exogenous Destruction of Capital As an example, an earthquake, or war, can destroy a significant chunk of existing capital. Then we must consider the transition path as the economy recovers. In particular, focus on Figure 1.4. Suppose we are in a steady-state k^* . All of a sudden the economy is hit by a shock which destroys a huge chunk of capital per-effective worker at time t_0 , leaving $k_{t_0} < k^*$. After t_0 , there are no more exogenous disturbances so the transition towards the original steady-state will be as if we began a world with $k_0 = k_{t_0}$. The reason the economy returns to the original steady-state is that there is nothing fundamentally different in the primitives due to the the exogenous shock.

A Change in the Savings Rate Sometimes we may be interested in what happens when the saving rate suddenly changes. In later chapters, the saving rate will be endogenous as households will decide optimally how much to save. But for now, it is useful to understand the effects of saving on capital accumulation and growth. In particular, suppose at time t_0 the saving rate increases. This does not affect the current stock of capital as k_{t_0} was determined by the saving rate at time $t_0 - 1$. However,

Figure 1.4: Dynamics of k after exogenous destruction of capital

the economy is fundamentally different in the sense that each period, a larger portion of output will be saved. Refer to Figure 1.5. Graphically, the entire function h is shifted upward proportionally which unambiguously results in a higher steady-state level $k^* > k^*$; you will verify this in Exercise 6. Transition to this new steady-state will still be monotone, and now follows a path described by the function h' , where h and h' differ only in terms of the saving rate parameter s and s' . In particular, the transition path will be as if the economy started at time t_0 with capital stock $k_0 = k^*$, with a law of motion described by h' .

Exercise 6. Use the Implicit Function Theorem to show that $\frac{\partial k^*}{\partial s} \in (0, 1)$.

Figure 1.5: Dynamics of k after a permanent increase in the saving rate

1.6 The Golden Rule of Capital Accumulation

Up to this point we have not discussed consumption. But households consume whatever they don't save. Hence, let C denote aggregate consumption, and $c = C/AL$ denote consumption per-effective worker. Then $c = (Y - S)/AL = (1 - s)f(k)$. So the dynamics of c are determined by output. Clearly, at any point in time, if more is saved then less is consumed. However, if more is saved today, then the stock of capital is higher tomorrow, and so is output. Then potentially, more can be consumed tomorrow. So it is interesting to ask whether a higher saving rate will lead to higher steady-state consumption or not.

Let us now focus on steady-state consumption c^* . It is easy to show that $c^* = f(k^*) - [(1 + n)(1 + g) - (1 - \delta)]k^*$. So what happens to steady-state consumption

when the saving rate is changed?

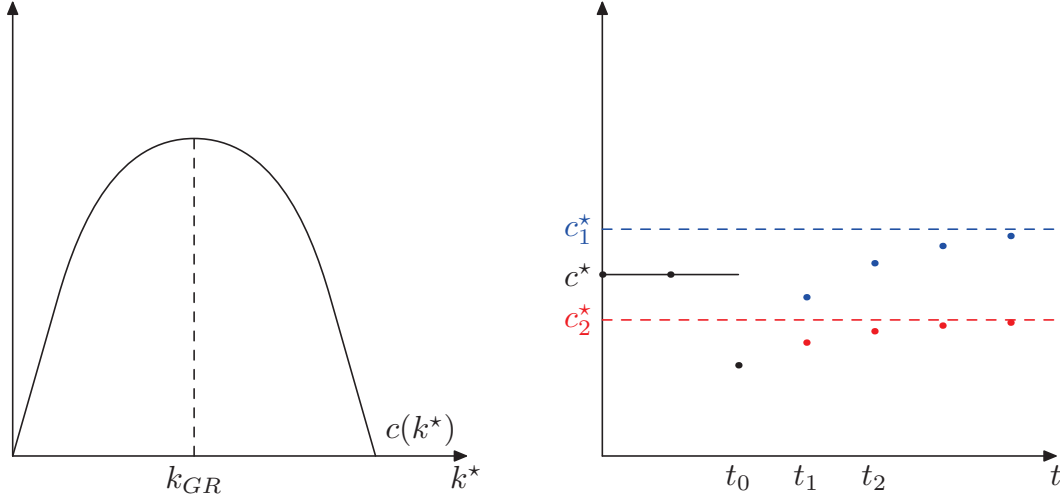
$$\frac{\partial c^*}{\partial s} = \underbrace{[f'(k^*) - (1+n)(1+g) - (1-\delta)]}_T \underbrace{\frac{\partial k^*}{\partial s}}_{+}$$

Note that the second term on the right hand side is positive. The intuition is that if more resources are saved each period (higher saving rate s), then capital will accumulate to a higher level. The first term, which we call T , is ambiguous. In fact there are three cases to consider.

- Case 1: $f'(k^*) > (1+n)(1+g) - (1-\delta)$, then $T > 0$ and steady-state consumption will increase.
- Case 2: $f'(k^*) < (1+n)(1+g) - (1-\delta)$, then $T < 0$ and steady-state consumption will decrease.
- Case 3: $f'(k^*) = (1+n)(1+g) - (1-\delta)$, then $T = 0$ and steady-state consumption will remain unchanged.

Case 3 defines the *Golden Rule* level of k^* , denoted by k_{GR} . This is the steady-state level of capital per-effective worker (among all possible steady-state levels) which maximizes steady-state consumption; see the left panel of Figure 1.6. Case one corresponds to when $k^* < k_{GR}$, and case two corresponds to when $k^* > k_{GR}$. Consider now the dynamics along the transition given an exogenous increase in s at say time $t = t_0$. Since k_{t_0} was determined at $t_0 - 1$, c_{t_0} unambiguously decreases in all cases. We know

Figure 1.6: Golden rule level of consumption (left), and consumption dynamics after a permanent increase in the saving rate (right)



Notes: $c(k^*)$ describes steady-state consumption per-effective worker given any steady-state capital per-effective worker k^* . In the right panel, c_1^* is the new steady-state consumption per-effective worker when $f'(k^*) > (1 + g)(1 + n) - (1 - \delta)$, and c_2^* is the new steady-state consumption per-effective worker when $f'(k^*) < (1 + g)(1 + n) - (1 - \delta)$.

that the path for consumption will converge monotonically towards its new steady-state since $c_t = (1 - s)f(k_t)$, and we already discussed the dynamics of k_t . See the right panel of Figure 1.6. The plot shows possible dynamics for cases one and two only. In case one, the new steady-state level of consumption per-effective worker is $c_1^* > c^*$, and in case two is $c_2^* < c^*$. Regarding the former, c_1^* corresponds to an original steady-state value for k_1^* to the left of k_{GR} , whereas in the latter case, c_1^* corresponds to an original steady-state value for k_1^* to the right of k_{GR} .

1.7 Speed of Convergence

Recall the law of motion equation (1.6). If we are in a neighborhood of the steady-state, then we can use a first order Taylor expansion to approximate the speed of convergence locally. Denote the dynamic map $k_t \mapsto k_{t+1}$ by G . We can approximate the dynamics in a neighborhood of the steady-state by using a first order Taylor expansion around the steady-state; that is,

$$k_{t+1} \approx k^* + G'(k_t)|_{k_t=k^*}(k_t - k^*) = k^* + G'(k^*)(k_t - k^*).$$

But we know that $G'(k) = \frac{sf'(k)+(1-\delta)}{(1+n)(1+g)}$. In addition, in the steady-state, it is the case that $s = \frac{k^*[(1+n)(1+g)-(1-\delta)]}{f(k^*)}$. Using these results we can rewrite the approximation as

$$k_{t+1} \approx k^* + \left[\frac{k^*[(1+n)(1+g)-(1-\delta)]f'(k^*)}{f(k^*)(1+n)(1+g)} + \frac{1-\delta}{(1+n)(1+g)} \right] (k_t - k^*).$$

In steady-state, the share of capital in output is $\alpha_K(k^*) = \frac{k^*f'(k^*)}{f(k^*)}$. So rewriting one more time we get

$$\begin{aligned} k_{t+1} &\approx k^* + \left[\alpha_K(k^*) + \frac{(1-\delta)(1-\alpha_K(k^*))}{(1+n)(1+g)} \right] (k_t - k^*) \\ &= \underbrace{\left[(1-\alpha_K(k^*)) - \frac{(1-\delta)(1-\alpha_K(k^*))}{(1+n)(1+g)} \right]}_b k^* \\ &\quad + \underbrace{\left[\alpha_K(k^*) + \frac{(1-\delta)(1-\alpha_K(k^*))}{(1+n)(1+g)} \right]}_a k_t. \end{aligned} \tag{1.9}$$

Note that terms are grouped so that a and b are just numbers, leaving a linear first order difference equation of the form $k_{t+1} = b + ak_t$. By our assumptions, it follows that $0 < a < 1$, and therefore a unique steady-state exists and is $k^* = \frac{b}{1-a}$.

Exercise 7. Consider a linear, first order, non-homogeneous difference equation of the form $x_{t+1} = ax_t + b$, with $x_0 > 0$. Show that the solution is $x_t = (x_0 - x^*)a^t + x^*$, with $x^* = b/(1 - a)$.

The solution to the difference equation (1.9) is therefore

$$k_t = [k_0 - k^*]a^t + k^*. \quad (1.10)$$

The term a determines how fast we converge, and since it is positive, it says that convergence is monotone from any $k_0 > 0$. In particular, if a is close to 1, then convergence is very slow. Conversely, if a is close to 0, then convergence is fast.

To be more precise about how fast the economy approaches a steady-state, we use the concept of *half-life*, which indicates how long it takes to get *half way* to k^* starting from k_0 . Note that we can not just measure how long it takes to *reach* k^* , since this point is never actually reached, unless of course $k_0 = k^*$; that is, the notion of steady-state, or any balanced growth path for that matter, is only asymptotic. So calculating the half-life amounts to finding the time $\hat{t}$ such that

$$\frac{k_{\hat{t}} - k^*}{k_0 - k^*} = \frac{1}{2}.$$

But we know $k_t = (k_0 - k^*)a^t + k^*$ which implies that $\frac{k_t - k^*}{k_0 - k^*} = a^t$, and consequently, $a^{\hat{t}} = \frac{1}{2}$ which means

$$\hat{t} = \frac{\ln(.5)}{\ln(a)}. \quad (1.11)$$

Example 2. Recall that when the production function is Cobb-Douglas, the number a is pinned down by (α, n, g, δ) . Set these to $(1/3, .01, .02, .06)$ respectively. Then we get $\hat{t} \approx 11.5$. That is, about 11 years and a half. If we change α to $2/3$, then we get $\hat{t} \approx 23.4$ about seventeen and a half years. So clearly, the speed of convergence depends crucially on α . Why?

1.8 Exercises

1. Suppose the technology displays constant returns to scale and has the Cobb Douglas form:

$$Y = K^{\alpha_1} (AL)^{\alpha_2} X^{1-\alpha_1-\alpha_2},$$

where X denotes the available quantity of the factor in fixed supply, with $X_t = \bar{X}$ all t . Population or raw labor (L) and labor efficiency (A) grow at the rates $(1 + n)$ and $(1 + g)$, respectively.

1. What is the rate of growth of aggregate output along a balanced growth path?
2. Under what conditions there is long run growth in per-capita output? Explain.
3. From now onwards, assume that the conditions that guarantee long-run growth in output per capita hold.

Note that the growth rate derived in the first point implicitly defines the variable needed to construct a normalized capital stock variable, $\tilde{k}$. That is, the variable Z so that $\tilde{k} = K/Z$.

Find the law of motion that governs the dynamics of $\tilde{k}$. Show that for any $\tilde{k} > 0$, this variable converges monotonically to a steady state value, and thus a unique balanced growth path exists.

4. Evaluate the following assertion: “An economy with a higher stock of a fixed resource will have a higher capital to output ratio in the long run”. Explain.
2. Consider now the following version of the Solow Model, where the production technology requires two distinct types of capital, K_1 and K_2 . The single good in the economy is produced via

$$Y = \bar{A}K_1^\alpha K_2^\beta (L)^{1-\alpha-\beta}.$$

Suppose for simplicity that there is no growth in labor efficiency. Population grows at the rate n . The parameter $\bar{A} > 0$ stands for the level of Total Factor Productivity.

As in the standard Solow model, there are constant saving rates for both types of capital. Let s_1 and s_2 be the exogenous saving rates for capital of types 1 and 2. Both types of capital depreciate at the same rate, $\delta \in (0, 1]$.

1. Derive the laws of motion for capital per worker of both types.
2. Solve explicitly for the steady-state values of capital per worker of both types.
3. Solve for the steady-state level of output per worker.

What is the *elasticity* of output per worker in steady state with respect to changes in Total Factor Productivity? That is, $\frac{d \log y}{d \log \bar{A}}$.

4. Consider now the standard growth model with Cobb-Douglas technology and capital share α . If $\alpha = \alpha_1 + \alpha_2$, is the elasticity in the standard case the same as in the case with two types of capital? Explain.